$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#2}_{0^+} \\ \hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{9}\left(Y_1 k^2 + 9\overset{(2)}{\kappa}_{2}\right) & \frac{1}{3}\left(Y_2 k^2 + 3\overset{(2)}{\kappa}_{2}\right) \\
\mathcal{K}^{\#2}_{0^+}{}^{\dagger} & \frac{1}{3}\left(Y_2 k^2 + 3\overset{(2)}{\kappa}_{2}\right) & \frac{1}{9}\left(Y_1 k^2 + 9\overset{(2)}{\kappa}_{2}\right)
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\ \hline
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{1}{9}\left(Y_3 k^2 + 3\overset{(2)}{\kappa}_{2}\right) & \frac{1}{9}\left(\sqrt{5}\,Y_3 k^2 + 3\sqrt{5}\,\overset{(2)}{\kappa}_{2}\right) \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{1}{9}\left(\sqrt{5}\,Y_3 k^2 + 3\sqrt{5}\,\overset{(2)}{\kappa}_{2}\right) & \frac{1}{9}\left(5\,Y_3 k^2 + 15\,\overset{(2)}{\kappa}_{2}\right)
\end{array}
$$

$$
\begin{array}{c|c}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} \\ \hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & -\frac{2k^2\overset{(4)}{\kappa}_{3}}{9}
\end{array}
\qquad
\begin{array}{c|c}
 & \mathcal{K}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\ \hline
\mathcal{K}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & -\frac{5k^2\overset{(4)}{\kappa}_{3}}{9}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#2}_{0^+} \\ \hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{Y_1 k^2+9\overset{(2)}{\kappa}_{2}}{9\,\mathrm{Det}(0^+)} & \frac{-Y_2 k^2-3\overset{(2)}{\kappa}_{2}}{3\,\mathrm{Det}(0^+)} \\
\mathcal{J}^{\#2}_{0^+}{}^{\dagger} & \frac{-Y_2 k^2-3\overset{(2)}{\kappa}_{2}}{3\,\mathrm{Det}(0^+)} & \frac{Y_1 k^2+9\overset{(2)}{\kappa}_{2}}{9\,\mathrm{Det}(0^+)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\ \hline
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{1}{6\,\mathrm{Det}(1^-)} & \frac{\sqrt{5}}{6\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{\sqrt{5}}{6\,\mathrm{Det}(1^-)} & \frac{5}{6\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|c}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} \\ \hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & -\frac{9}{2k^2\overset{(4)}{\kappa}_{3}}
\end{array}
\qquad
\begin{array}{c|c}
 & \mathcal{J}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\ \hline
\mathcal{J}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & -\frac{9}{5k^2\overset{(4)}{\kappa}_{3}}
\end{array}
$$

Abbreviations used in matrices

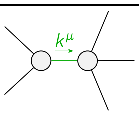
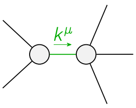
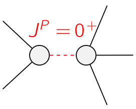
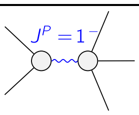
$$Y_1 == 9\left(\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2}\right) - 2\overset{(4)}{\kappa}_{3} \;\&\&\; Y_2 == 3\left(\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2}\right) - \overset{(4)}{\kappa}_{3} \;\&\&\; Y_3 == 3\overset{(4)}{\kappa}_{2} - \overset{(4)}{\kappa}_{3} \;\&\&$$

$$\mathrm{Det}(0^+) == \frac{1}{81}\left(k^4\left(18\left(\overset{(4)}{\kappa}_{1}+\overset{(4)}{\kappa}_{2}\right)-5\overset{(4)}{\kappa}_{3}\right)\overset{(4)}{\kappa}_{3} + 18k^2\overset{(4)}{\kappa}_{3}\overset{(2)}{\kappa}_{2}\right) \;\&\&\; \mathrm{Det}(1^-) == k^2\left(2\overset{(4)}{\kappa}_{2} - \frac{2\overset{(4)}{\kappa}_{3}}{3}\right) + 2\overset{(2)}{\kappa}_{2}$$

Lagrangian

$$\overset{(2)}{\kappa}_{2}\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}_{\beta}{}^{\chi}{}_{\chi} + \overset{(4)}{\kappa}_{1}\,\partial_{\beta}\mathcal{K}_{\chi}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{2}\,\partial_{\chi}\mathcal{K}_{\beta}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} +$$

$$\overset{(4)}{\kappa}_{3}\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{2}{3}\overset{(4)}{\kappa}_{3}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{5}{9}\overset{(4)}{\kappa}_{3}\,\partial_{\delta}\mathcal{K}_{\alpha\beta\chi}\,\partial^{\delta}\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}^{\#2}_{1^-}{}^{\alpha} == 5\,\mathcal{J}^{\#1}_{1^-}{}^{\alpha}$	3	$6\,\partial_{\delta}\partial_{\chi}\partial_{\beta}\partial^{\alpha}\mathcal{J}^{\beta\chi\delta} + \partial_{\delta}\partial^{\delta}\partial_{\chi}\partial^{\chi}\mathcal{J}^{\alpha\beta}{}_{\beta} == \partial_{\delta}\partial^{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + 6\,\partial_{\delta}\partial^{\delta}\partial_{\chi}\partial_{\beta}\mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	3		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\overset{(4)}{\kappa}_{3}}$
	4	0	$\frac{1}{\overset{(4)}{\kappa}_{3}}$
	1	$-\frac{18\overset{(2)}{\kappa}_{2}}{18\left(\overset{(4)}{\kappa}_{1}+\overset{(4)}{\kappa}_{2}\right)-5\overset{(4)}{\kappa}_{3}}$	$\frac{9}{18\overset{(4)}{\kappa}_{1}+18\overset{(4)}{\kappa}_{2}-5\overset{(4)}{\kappa}_{3}}$
	3	$\frac{3\overset{(2)}{\kappa}_{2}}{-3\overset{(4)}{\kappa}_{2}+\overset{(4)}{\kappa}_{3}}$	$-\frac{3}{6\overset{(4)}{\kappa}_{2}-2\overset{(4)}{\kappa}_{3}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		